

Raunak Kumar

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SUMMARY I am passionate about algorithms, machine learning, systems and competitive programming. I take keen interest in machine learning research and teaching.

EDUCATION *B.Sc. Computer Science,* September 2013 - May 2018
University of British Columbia, Vancouver, BC

- Grade in Computer Science and Mathematics courses: 90.4%
- Related Coursework:
 - Machine Learning (graduate course), Advanced Algorithm Design and Analysis, Randomized Big Data Algorithms (graduate course), Theory of Computation
 - Probability, Real Analysis, Graph Theory, Linear Algebra, Linear Programming
 - Internet Computing, Advanced Operating Systems, Databases

AWARDS J. Fred Muir Memorial Scholarship in Science ('13-'14)
Trek Excellence Scholarship ('13-'14) - Top 5% of undergraduates
Science Scholar / Dean's Honors List ('13-'14, '14-'15, '15-'16, '16-'17)
UBC Computer Science Teaching Assistant Award ('16-'17)

TECHNOLOGY SKILLS *Programming Languages:* C++, Python, C, Matlab, Java.
Software: TensorFlow, PyTorch, Numpy, Scikit-Learn, Vim, Git.
Operating Systems: Linux Ubuntu, Windows.

EXPERIENCE *Software Engineering Intern* May 2016 - August 2016
Google, Mountain View, CA

- Worked on a distributed datastore trigger utility in C++ for the App and Product Infrastructure team. Triggers are a mechanism by which actions can be scheduled for the future. The goal of the utility is to provide an easy and extensible way to maintain and debug triggers.

NSERC Research Assistantship May 2017 - Present
Computer Science Department, UBC

- Working on theoretical and applied machine learning problems under the supervision of Prof. Mark Schmidt. We are using recent developments in convex optimization to analyse the convergence rate of Expectation-Maximization in the case of mixture of Gaussians, and studying better curriculum learning and batch sampling strategies for neural networks.

TEACHING

Instructor / Coordinator

January 2017 - April 2017

Computer Science Department, UBC

- Co-taught CPSC 490: Problem Solving in Computer Science, focusing on practical applications of advanced algorithms and data structures. Responsibilities included lecturing a class of 15-30 students 3 times a week, writing lecture notes and writing assignments. Website: www.ugrad.cs.ubc.ca/~cs490/2016W2/.

Teaching Assistant

September 2014 - Present

Computer Science Department, UBC

- Worked as a TA for 8 terms for the following courses: Advanced Algorithm Design and Analysis, Machine Learning, Intermediate Algorithm Design and Analysis, Introduction to Computer Systems, Models of Computation. Responsibilities included lecturing sections ranging from 15 - 100 students, holding office hours and grading. I have taught an entire lecture for two algorithms courses as well. I was an undergrad when I worked as a TA for the Advanced Algorithms and Machine Learning courses. Usually, only grad students are assigned to TA these courses.

PROJECTS

Programming Competitions

- UBC ACM Team: 2015 - Present.
- ACM-ICPC PacNW Regionals: 15th (2016, Div. 1), 6th (2015, Div. 2).
- Calgary Collegiate Programming Competition: 3rd (2015, Div. 2).
- SFU Winter Programming Contest: 8th (2017).
- NAIPC (Open Division USA/Canada): 27th (2016).

Approximating Steiner Trees in the Semi-Streaming Model

- Presented a $2(1+\epsilon)(2\frac{\log n}{\log \log n} + 1)$ -approximation for the weighed metric Steiner tree problem using $O(n \text{ polylog } n)$ space, as part of a course project for a graduate class on randomized big data algorithms.

Kernel

- Implemented basic functionality of a kernel using C and x86 assembly on the Bochs emulator. Features included memory management, process management, system calls, interrupt handling, signals and hardware interrupts.

Web Proxy Server

- Created a multithreaded server application using C and Unix sockets to access HTTP content from websites, block blacklisted websites and cache results.

DNS Client

- Created a scaled down DNS resolver using Java and UDP datagrams which interacts with various DNS servers to resolve a domain name into an IP address.

Dictionary Client

- Created a client side application using Java and TCP to connect with specified dictionary servers and retrieve the requested definitions.

insightUBC

- Created a backed using Typescript and Node.js to query UBC rooms and courses data.

REFERENCES William Evans, email: will@cs.ubc.ca, phone: 604-822-0827
Mark Schmidt, email: schmidtm@cs.ubc.ca, phone: 604-822-6421
Nicholas Harvey, email: nickhar@cs.ubc.ca, phone: 604-822-5491

OTHER Canadian citizen.